

Exam 1

● Graded

Student

Veer Shukla

Total Points

97 / 100 pts

Question 1

(no title)

14 / 14 pts

1.1 — (no title)

2 / 2 pts

✓ - 0 pts Correct. The correct answer was False.

- 2 pts Incorrect. The correct answer was False.

- 1 pt Failure to follow instructions.

- 2 pts Answer unclear.

- 2 pts Blank.

1.2 — (no title)

2 / 2 pts

✓ - 0 pts Correct.

- 2 pts Incorrect.

- 1 pt Failure to follow instructions.

- 2 pts Answer unclear.

- 2 pts Blank.

1.3 — (no title)

2 / 2 pts

✓ - 0 pts Correct. The correct answer is False.

- 2 pts Incorrect. The correct answer was False.

- 1 pt Failure to follow instructions.

- 2 pts Answer unclear.

- 2 pts Blank.

1.4 — (no title)

2 / 2 pts

✓ - 0 pts Correct. The correct answer is True.

- 2 pts Incorrect. The correct answer was True.

- 1 pt Failure to follow instructions.

- 2 pts Answer unclear.

- 2 pts Blank.

1.5 — (no title)

2 / 2 pts

✓ - 0 pts Correct. The correct answer is False.

- 2 pts Incorrect. The correct answer was False.

- 1 pt Failure to follow instructions.

- 2 pts Answer unclear.

- 2 pts Blank.

1.6 (no title) 2 / 2 pts

✓ - 0 pts Correct.

- 2 pts Incorrect.

- 1 pt Failure to follow instructions.

- 2 pts Answer unclear.

- 2 pts Blank.

1.7 (no title) 2 / 2 pts

✓ - 0 pts Correct. The correct answer is False.

- 2 pts Incorrect. The correct answer was False.

- 1 pt Failure to follow instructions.

- 2 pts Answer unclear.

- 2 pts Blank.

Question 2

(no title) 4 / 6 pts

- 0 pts Correct

- 2 pts Error related to original denominator factor x^2 .

✓ - 2 pts Error related to original repeated linear factor in denominator For example: Numerator of linear factors must be constants, not linear functions.

- 2 pts Error related to original irreducible quadratic factor in denominator. For example: Numerator of irreducible quadratic factors must be linear.

- 6 pts Blank.

- 0.5 pts Algebraic error. Form related to x^2 factor should look like $\frac{C_1}{x} + \frac{C_2}{x^2}$. Note: x^2 is not an irreducible quadratic factor.

- 1.25 pts Algebraic error. Form related to (linear)³ factor should look like $\frac{B_1}{\text{linear}} + \frac{B_2}{\text{linear}^2} + \frac{B_3}{\text{linear}^3}$.

- 2 pts Algebraic error. Form related to (quadratic)² factor should look like $\frac{Fx+G}{\text{quadratic}} + \frac{Hx+I}{\text{quadratic}^2}$.

- 0.2 pts Other error.

- 6 pts No partial credit possible.

- 0.75 pts (minor) Missing term.

- 1.5 pts (major) Missing lower degree repeated term.

- 1.5 pts Exponents in denominators neglected.

✓ - 0 pts Correct

- 7 pts Part a). Integration by parts is not implemented correctly
- 2 pts Part a) Algebraic simplification is wrong
- 7 pts trig substitution is correct but implementation of it is not complete or wrong
- 5 pts Part b) Integration has multiple mistakes
- 3 pts Part a) no inverse substitution or inverse substitution is wrong
- 7 pts Part b) Substitution is not correct or the implementation of the correct substitution is wrong
- 10 pts Part b) is wrong or no work to grade
- 10 pts Part a) is wrong or no work to grade
- 2 pts Part b) Not major integration mistake - not correct the numerical coefficient
- 8 pts Part a) integration is not correct
- 5 pts After the original integral split into two parts, one of the integrals was not taken correctly or was not taken at all
- 1 pt $\ln(\cos(\arctan(z/3)))$ not simplified.
- 1 pt Part a) There is no inverse substitution
- 5 pts Part a) Integration is not complete./not correct

✓ **+ 10 pts** Correct answer: $\frac{1}{3} \sin^3(\theta) - \frac{1}{5} \sin^5(\theta) + C$ | **If points are deducted, below READ the rubric CAREFULLY for the explanations.** YOUR REQUEST FOR A REGRADE WILL BE TAKEN CARE OF ONLY IF YOUR INSTRUCTOR APPROVES IT. WITHOUT THE APPROVAL NOTHING WILL BE REGRADED.

- **1 pt** Your work is missing $+C$ once | CAREFULLY GO OVER YOUR WORK. | $+C$ must be stated ANY TIME integration was done. When you integrated in terms of u , you didn't state $+C$
- **2 pts** Your work is missing $+C$ after integration
- **10 pts** completely wrong and/or not finished | Correct answer: $\frac{1}{3} \sin^3(\theta) - \frac{1}{5} \sin^5(\theta) + C$
- **3 pts** You don't know what is the derivative of sine. Your answer is incorrect | Correct answer: $\frac{1}{3} \sin^3(\theta) - \frac{1}{5} \sin^5(\theta) + C$
- **1 pt** Check the signs / exponents | Correct answer: $\frac{1}{3} \sin^3(\theta) - \frac{1}{5} \sin^5(\theta) + C$
- **2 pts** Kept the integral sign after integration was done, check your work carefully.
- **3 pts** Incorrect trigonometric identity used. Your work has incorrect signs. Your answer is incorrect.
- **5 pts** you didn't apply trigonometric identity $\cos^2(\theta) = 1 - \sin^2(\theta)$ correctly or not at all, so your work is incorrect. | Correct answer: $\frac{1}{3} \sin^3(\theta) - \frac{1}{5} \sin^5(\theta) + C$
- **4 pts** Integration is done incorrectly. Check your work. | Correct answer: $\frac{1}{3} \sin^3(\theta) - \frac{1}{5} \sin^5(\theta) + C$
- **3 pts** you don't know what is the derivative of cosine.
- **8 pts** mostly wrong steps | Correct answer: $\frac{1}{3} \sin^3(\theta) - \frac{1}{5} \sin^5(\theta) + C$
- **2 pts** A small algebraic mistake in the middle of your work. | Correct answer: $\frac{1}{3} \sin^3(\theta) - \frac{1}{5} \sin^5(\theta) + C$
- **2 pts** Your work is missing appropriate usage of the integral signs. You need to write $\int$ every time before integration is done.
- **3 pts** Your final answer is missing θ 's under sines. Also you didn't integral u^4 correctly. Your work has mistakes, | Correct answer: $\frac{1}{3} \sin^3(\theta) - \frac{1}{5} \sin^5(\theta) + C$
- **2 pts** | Correct answer: $\frac{1}{3} \sin^3(\theta) - \frac{1}{5} \sin^5(\theta) + C$
- **7 pts** You didn't show work for I_1 and for I_2 at all. Your method is not efficient. It was supposed to be done by u -substitution. | Correct answer: $\frac{1}{3} \sin^3(\theta) - \frac{1}{5} \sin^5(\theta) + C$
- **2 pts** substitution is not shown at all
- **8 pts** Incorrect substitution. Answer: $\frac{1}{5} \cos^5(\theta) - \frac{1}{3} \cos^3(\theta) + C$
- **10 pts** This integral should be done by u -substitution. Your work is incorrect and not finished. Answer: $\frac{1}{5} \cos^5(\theta) - \frac{1}{3} \cos^3(\theta) + C$

✓ + 10 pts Correct answer $\frac{3}{4}$ | **If points are deducted, below READ the rubric CAREFULLY for the explanations.** YOUR REQUEST FOR A REGRADE WILL BE TAKEN CARE OF ONLY IF YOUR INSTRUCTOR APPROVES IT. WITHOUT THE APPROVAL NOTHING WILL BE REGRADED.

- 7 pts Mostly incorrect. After integrations you were supposed to get $-\frac{1}{2}(x+1)\cos(2x) + \frac{1}{4}\sin(2x)|_0^{\pi/4}$, and after evaluating it, the answer should be $\frac{3}{4}$
- 3 pts After evaluating the integral, the answer should be $\frac{3}{4}$. Check your work.
- 1 pt Mistake when you integrated $\cos(2x)$ OR $\sin(2x)$. Check your work.
- 10 pts all work is incorrect. You had to clearly show integration by parts where all parts are labeled. Nothing is clearly shown. | Answer: $\frac{3}{4}$
- 4 pts Forgot to evaluate the integral. Check the signs in your final result, it is incorrect. | Answer: $-\frac{1}{2}(x + 1)\cos(2x) + \frac{1}{4}\sin(2x)|_0^{\pi/4} = \frac{3}{4}$
- 9 pts not finished
- 1 pt inappropriately used $+C$ OR missing $+C$ when it is required
- 1 pt check the signs
- 1 pt Answer: $\frac{3}{4}$
- 2 pts Work is not presented in a clear and appropriate way. Next time I will deduct more points for this kind of work.

Question 5

(no title)

20 / 20 pts

✓ - 0 pts Correct

- 1 pt Need absolute value or constant.
- 1 pt small errors like copying error, not simplified, etc
- 10 pts integration missing or completely wrong
- 18 pts mostly wrong
- 20 pts Click here to replace this description.
- 2 pts algebra error
- 8 pts Big error in finding the partial fraction decomposition
- 16 pts The partial fraction decomposition part, is completely wrong.
- 2 pts errors, or bad writing, mathematically completely wrong.
- 4 pts integration error; or other error
- 7 pts major mistake after finding the decomposition.
- 5 pts several errors

Question 6

(no title)

19 / 20 pts

✓ + 20 pts all correct

- + 11 pts Got to $\int \tan^2(\theta) d\theta$ correctly
- + 10 pts Did not do the correct substitution, but the rest of the integral was done correctly (starting from an incorrect place).
- + 5 pts Correct choice of substitution $x = 2\sec(\theta)$, but did not do the substitution correctly
- + 3 pts Tried a trig substitution, but didn't get very far.
- + 1 pt tried some other substitution

✓ - 1 pt off by a factor of 2 or off by minus sign

- 1 pt mistake going back to the original variable
- 2 pts couple mistakes going back to original variable
- 3 pts didn't go back to the original variable
- 5 pts Substitution $x = 2\sec(\theta)$ was correct, but did not do the substitution correctly, but the rest of the integral was done correctly, given the initial error
- 5 pts mostly correct but did not finish the integrals correctly
- 2 pts can't leave a composition of trig(inverse trig)
- + 0 pts Nothing worth credit

Question 7

(no title)

0 / 0 pts

7.1 (no title)

0 / 0 pts

– 0 pts This exam was 1/7 hard.

– 0 pts This exam was 2/7 hard.

– 0 pts This exam was 3/7 hard.

✓ – 0 pts This exam was reasonable.

– 0 pts This exam was 5/7 hard.

– 0 pts This exam was 6/7 hard.

– 0 pts This exam was 7/7 hard.

– 0.1 pts Blank.

– 0.1 pts Failure to follow instructions.

7.2 (no title)

0 / 0 pts

– 0 pts WebAssign helped 1/7.

– 0 pts WebAssign helped 2/7.

– 0 pts WebAssign helped 3/7.

– 0 pts WebAssign helped 4/7.

– 0 pts WebAssign helped 5/7.

✓ – 0 pts WebAssign helped 6/7.

– 0 pts WebAssign helped 7/7.

– 0.1 pts Blank.

– 0.1 pts Failure to follow instructions.

7.3 (no title)

0 / 0 pts

– 0 pts The time was 1/7.

– 0 pts The time was 2/7.

– 0 pts The time was 3/7.

✓ – 0 pts The time was 4/7.

– 0 pts The time was 5/7.

– 0 pts The time was 6/7.

– 0 pts The time was 7/7.

– 0.1 pts Blank.

– 0.1 pts Failure to follow instructions.

- 0 pts The exam required pure creativity.
- 0 pts The exam required mostly creativity (2/7).
- 0 pts The exam required more creativity (3/7).
- 0 pts The exam equally required creativity and computation.

✓ - 0 pts The exam was more computational (5/7).

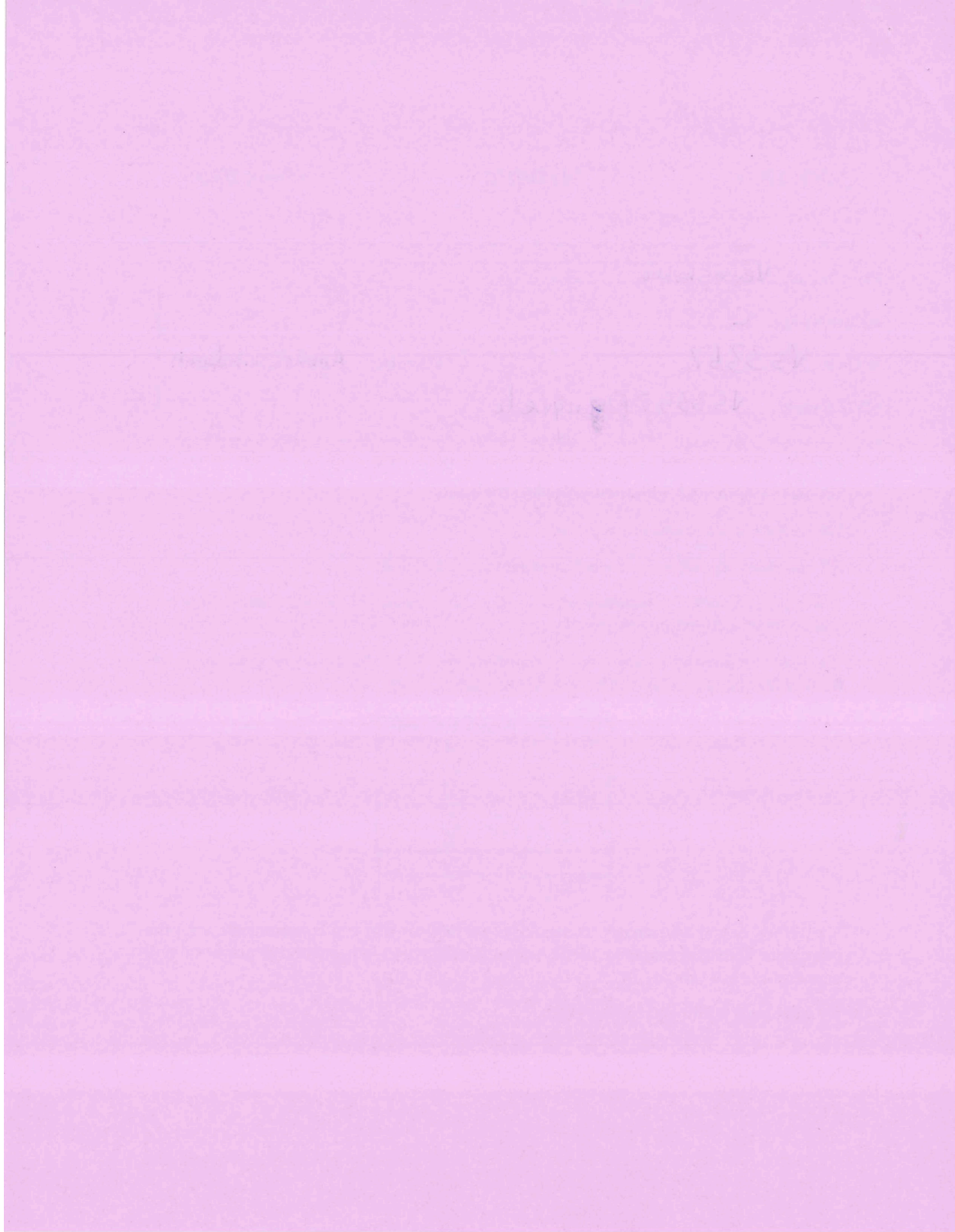
- 0 pts The exam was mostly computational (6/7).
- 0 pts The exam was all computation.
- 0.1 pts Blank.
- 0.1 pts Failure to follow instructions.

Print Name: VeegshwlaSection Letter: CN-ID #: VS3357Instructor: Amakoe GbedemahNYU E-mail: VS3357@nyu.edu

- You have 1 hour + 20 minutes to complete the exam.
- This exam is graded out of 100 points.
- All questions should include work in order to receive credit.
- You will lose points if you leave your answer as a composition of a trigonometric function and an inverse trigonometric function.
- You may use the back of any page to continue your work. If you do so, indicate on the front of that page in order for the graders to know to look.

Problem	Points Available
1	14
2	6
3	20
4	20
5	20
6	20
Total	100

- No cell phones, iPads, smart watches, or other devices that can communicate with the internet or with others may be used. Any such equipment found with the power on will be interpreted as *cheating*.
- **No calculator** is allowed for this exam.



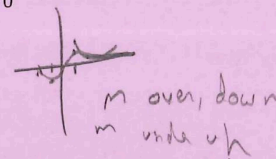
1. For each of the following, fill in one circle (● not ✓ nor ×) corresponding to *True* or *False*. You do not need to show work; you will not receive partial credit. Answers written in the margin will be not be graded. (2 points each)

True False

- (a) ☐ ☒ A left-endpoint approximation will overestimate the value of $\int_0^1 x^2 dx$.



- (b) ☒ ☐ A midpoint approximation of $\int_1^4 \frac{1}{x} dx$ is an underestimate.



- (c) ☐ ☒ A trapezoidal approximation of $\int_1^4 \frac{1}{x} dx$ will underestimate the value.



- (d) ☒ ☐ A right-endpoint approximation will overestimate the value $\int_{-10}^{15} e^x dx$.

(e) ☐ ☒ $\frac{d}{dx} \int_{-x}^{2x} \cos(t^3) dt = \cos(8x^3) - \cos(x^3)$

$\frac{d}{dn} \int_{-n}^{2n} f(t) dt \rightarrow \frac{d}{dn} [f(2n) - f(-n)]$
 $2f(2n) + f(-n)$
 $2 \cos(8n^3) + \cos(n^3)$

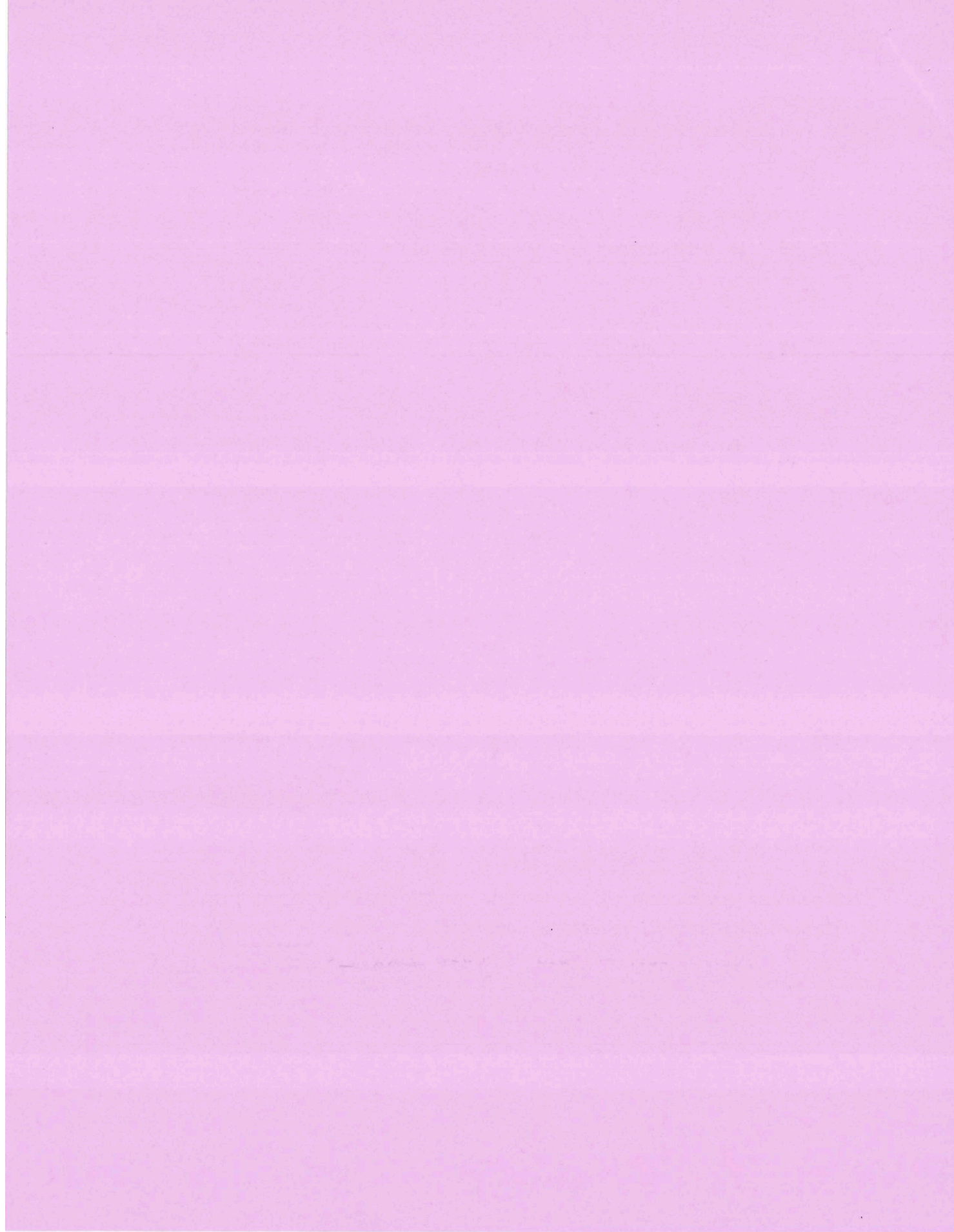
- (f) ☒ ☐ $\int_{2x}^{x^3} \frac{d}{dz} (\sin(z^2)) dz = \sin(x^6) - \sin(4x^2)$

- (g) ☐ ☒ $\frac{x^2 + 3x}{x^2 - 1} = \frac{1}{x+1} + \frac{2}{x-1}$
- $2n+2+x-1$
 $3n$

2. (6 points) Determine the *form* of the partial fractions decomposition of $\frac{x^8 - 15x^6 + 3x^3 - 45}{x^2(2x+1)^3(3+x^2)^2}$.

Do not find the numerical values of the numerators.

$$\frac{A}{n} + \frac{B}{n^2} + \frac{Cn+D}{(2n+1)^3} + \frac{En+F}{(2n+1)^2} + \frac{Gn+H}{(2n+1)} + \frac{Ix+J}{(3+x^2)} + \frac{Kx+L}{(3+x^2)^2}$$



The remaining questions must include all work.

3. Find the indicated antiderivatives. Show all work. (10 points each)

(a) $\int (u-4)^{2025} (u+3) du$ $t = u-4,$
 $\frac{dt}{du} = 1, dt = du$

$$\int t^{2025} (t+7) dt$$

$$\int (t^{2026} + 7t^{2025}) dt$$

$$\frac{t^{2027}}{2027} + 7 \frac{t^{2026}}{2026} + C$$

$$\boxed{\frac{(u-4)^{2027}}{2027} + \frac{7(u-4)^{2026}}{2026} + C = \int (u-4)^{2025} (u+3) du}$$

(b) $\int \frac{3+2z}{9+z^2} dz$

$$\int \frac{3}{9+z^2} dz + \int \frac{2z}{9+z^2} dz$$

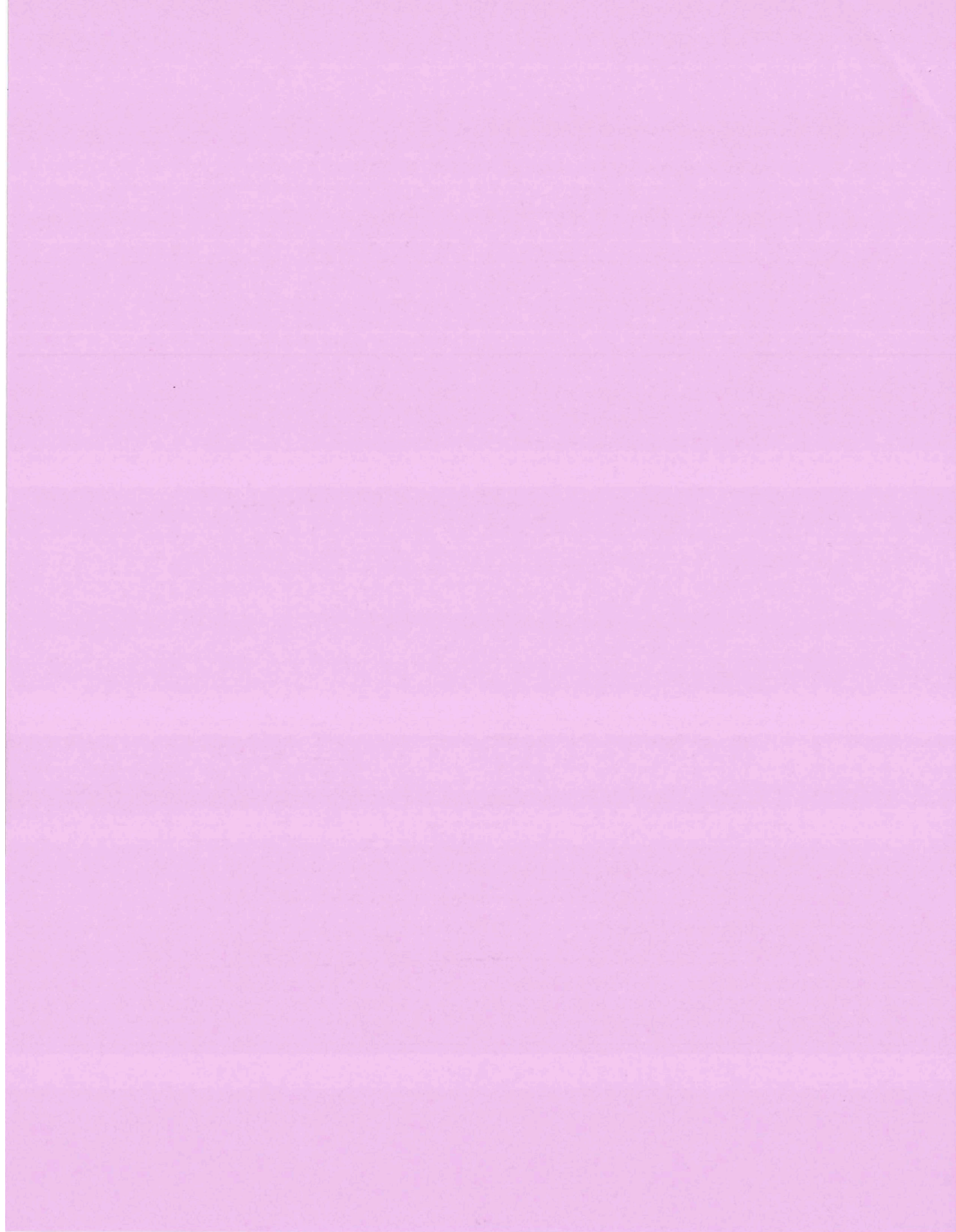
$$u = 9+z^2$$

$$\frac{du}{dz} = 2z$$

$$3 \int \frac{1}{3^2+z^2} dz + \int \frac{\cancel{2z} \times du}{u \cancel{2z}}$$

$$3 \times \frac{1}{3} \arctan\left(\frac{z}{3}\right) + \ln|u|$$

$$\boxed{\arctan\left(\frac{z}{3}\right) + \ln|9+z^2| + C = \int \frac{3+2z}{9+z^2} dz}$$



4. Show all work. (10 points each)

(a) Find the antiderivative. $\int \sin^2 \theta \cos^3 \theta d\theta$

$$\int \sin^2(\theta) \cos^3(\theta) d\theta$$

$$\int \sin^2(\theta) \cos^2(\theta) \cos(\theta) d\theta$$

$$\int \sin^2(\theta) (1 - \sin^2(\theta)) \cos(\theta) d\theta$$

$$\boxed{U = \sin(\theta)}, du = \cos \theta d\theta$$

$$\int u^2 (1 - u^2) du$$

$$\int (u^2 - u^4) du \rightarrow \frac{u^3}{3} - \frac{u^5}{5} + C$$

$$\boxed{\int \sin^2(\theta) \cos^3(\theta) d\theta = \frac{\sin^3(\theta)}{3} - \frac{\sin^5(\theta)}{5} + C}$$

(b) Evaluate the integral (and simplify). $\int_0^{\pi/4} (x+1) \sin(2x) dx$

$$\int u dv = UV - \int v du$$

$$\int_0^{\pi/4} (x+1) \sin(2x) dx$$

$$U = x+1,$$

$$v = -\frac{1}{2} \cos(2x)$$

$$du = dx$$

$$dv = \sin(2x) dx$$

$$- \frac{(x+1)}{2} \cos(2x) + \frac{1}{2} \int \cos(2x) dx$$

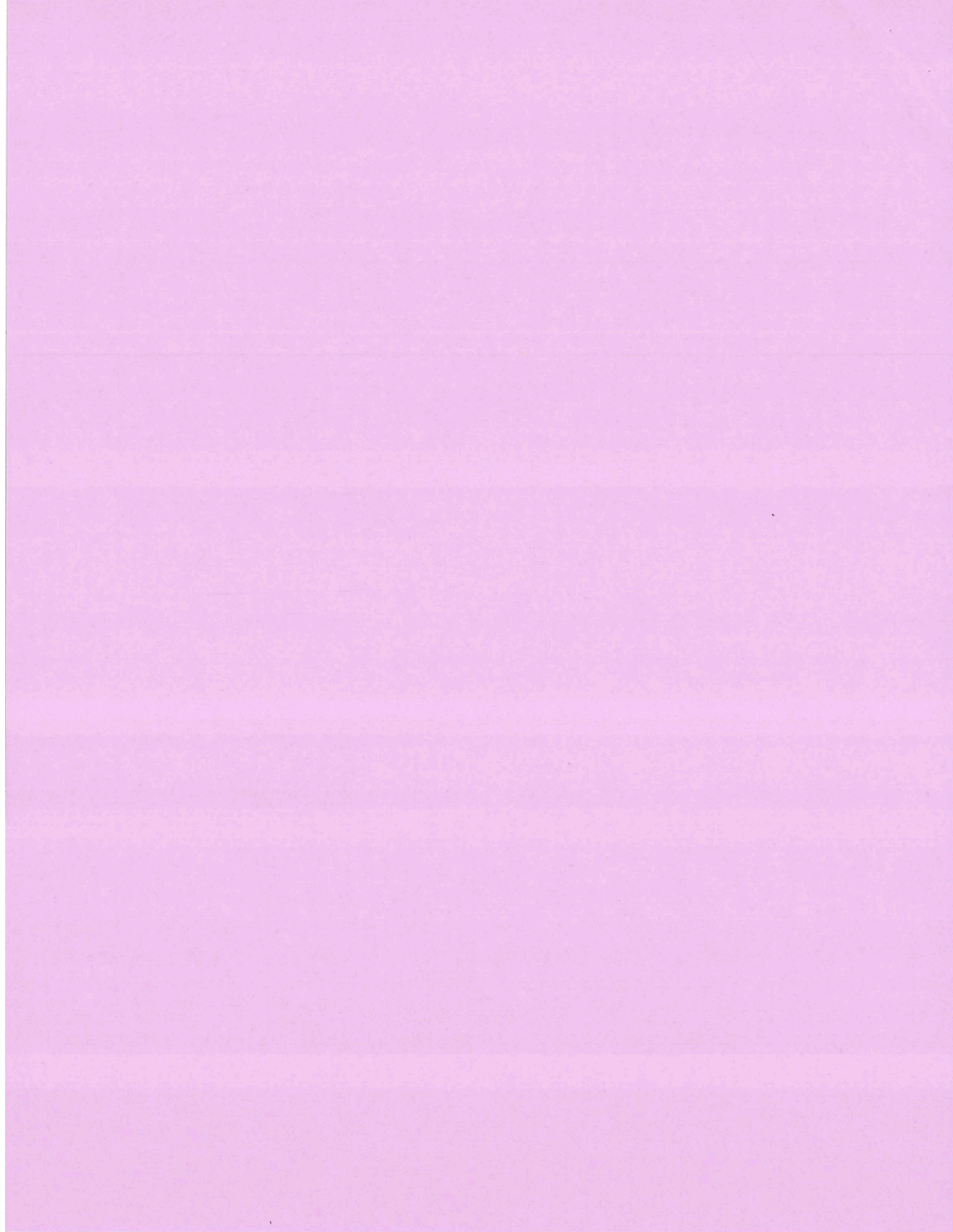
$$- \frac{(x+1)}{2} \cos(2x) + \frac{1}{2} \times \frac{1}{2} \sin(2x)$$

$$\left[- \frac{(x+1)}{2} \cos(2x) + \frac{1}{4} \sin(2x) \right]_0^{\pi/4}$$

$$\left[- \frac{1}{2} \left(\frac{\pi}{4} + 1 \right) \cos\left(\frac{\pi}{2}\right) + \frac{1}{4} \sin\left(\frac{\pi}{2}\right) \right] - \left[- \frac{1}{2} \cos(0) + \frac{1}{4} \sin(0) \right]$$

$$\left[- \frac{1}{2} \left(\frac{\pi+4}{4} \right) \times 0 + \frac{1}{4} \right] - \left[- \frac{1}{2} + 0 \right]$$

$$\frac{1}{4} + \frac{1 \times 2}{2 \times 2} \boxed{\frac{3}{4}}$$



5. Find the indicated antiderivative. Show all work. (20 points)

$$\int \frac{3x+1}{x^2-3x} dx$$

$$\int \frac{3n+1}{n(n-3)} dn$$

$$\frac{A}{n} + \frac{B}{n-3}$$

$$A(n-3) + B(n) = 3n+1$$

$$An - 3A + Bn = 3n+1$$

$$(A+B)n - 3A = 3n+1$$

$$-3A = 1$$

$$A = -\frac{1}{3}$$

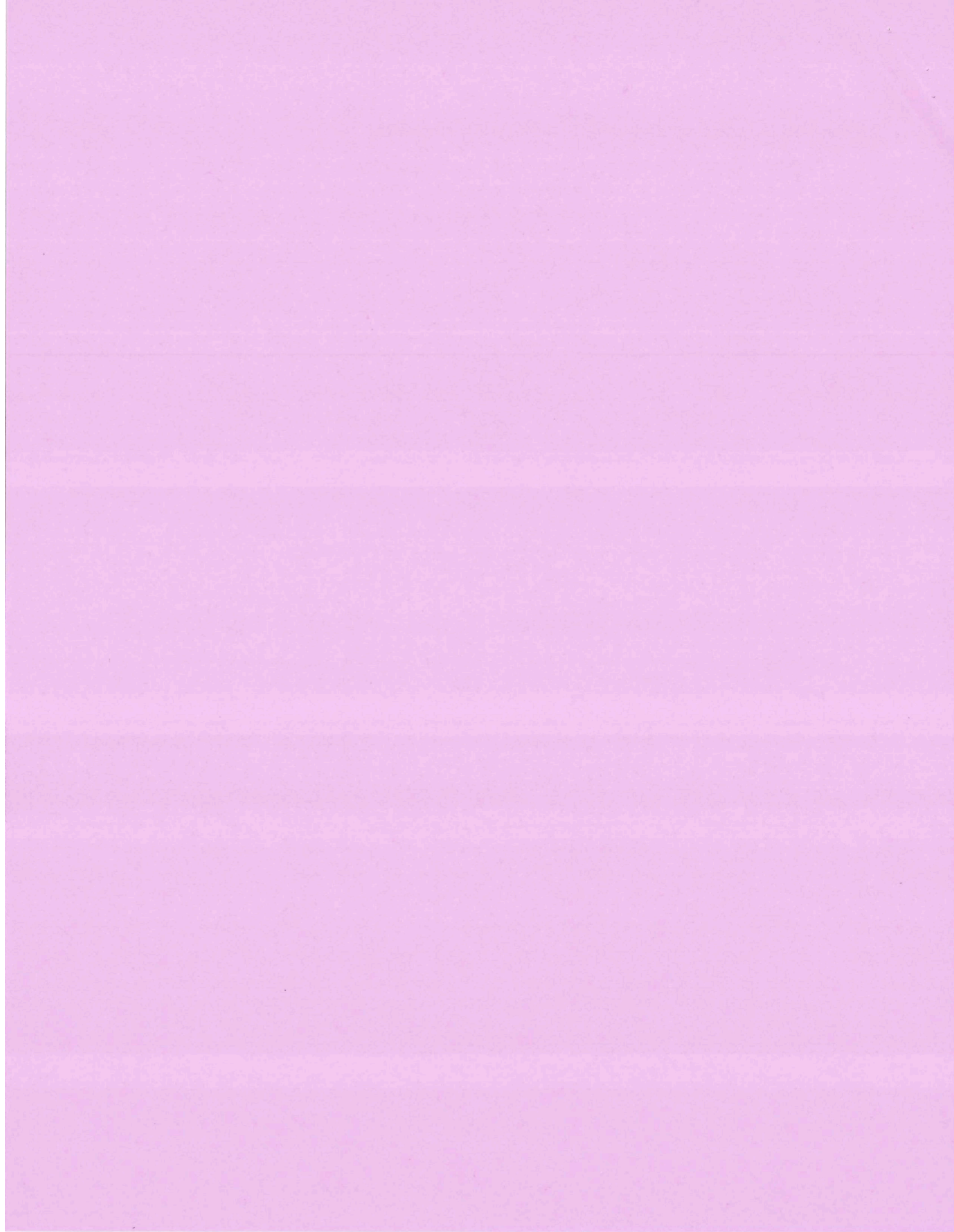
$$-\frac{1}{3} + B = 3$$

$$B = 3 + \frac{1}{3} = \frac{10}{3}$$

$$\int \frac{10}{3(n-3)} - \frac{1}{3n}$$

$$\frac{10}{3} \int \frac{1}{n-3} dn - \frac{1}{3} \int \frac{1}{n} dn$$

$$\int \frac{3n+1}{n^2-3n} dn = \frac{10}{3} \ln|n-3| - \frac{1}{3} \ln|n| + C$$



$$\begin{aligned} &\sqrt{a^2+x^2} \\ &\sqrt{a^2+u^2} \tan \theta \\ &\sqrt{u^2-a^2} \end{aligned}$$

6. Find the indicated antiderivative. Show all work. (20 points)

$$\int \frac{\sqrt{x^2-4}}{x} dx$$

$$\sqrt{x^2-a^2}$$

$$x = 2 \sec \theta$$

$$dx = 2 \sec \theta \tan \theta d\theta$$

$$\int \frac{\sqrt{4 \sec^2(\theta) - 4}}{2 \sec(\theta)} \times 2 \sec(\theta) \tan(\theta) d\theta$$

$$\int \sqrt{2^2 (\sec^2(\theta) - 1)} \tan(\theta) d\theta$$

$$\int 2 \tan^2 \theta d\theta$$

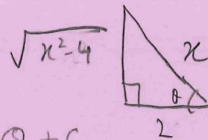
$$2 \int \tan^2 \theta d\theta$$

$$2 \int (\sec^2(\theta) - 1) d\theta$$

$$2 \int \sec^2(\theta) d\theta - \int d\theta$$

$$2 \tan(\theta) - \theta + C$$

$$\frac{x}{2} = \sec(\theta) = \frac{1}{\cos(\theta)}$$



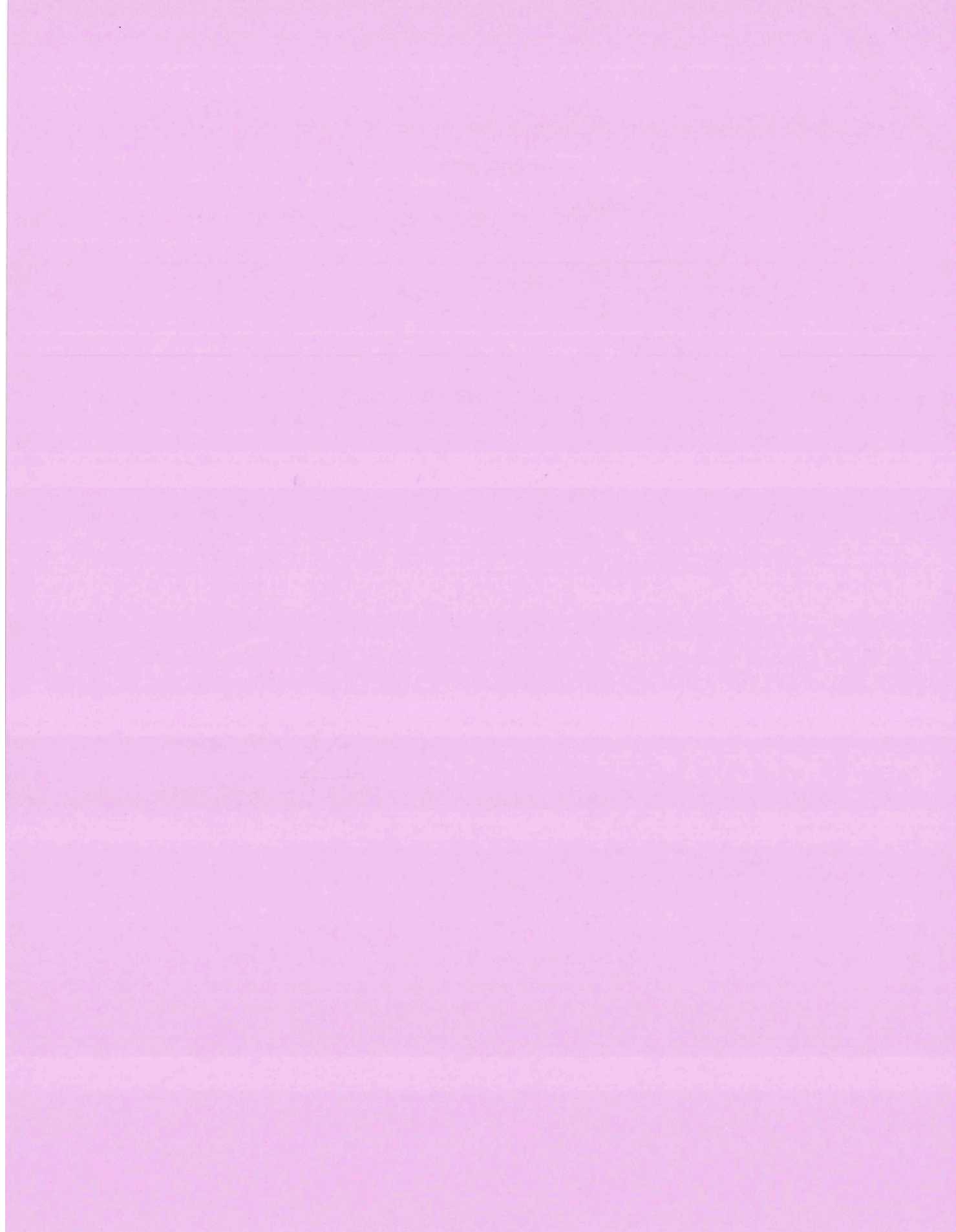
$$\frac{2}{x} = \cos(\theta)$$

$$\arccos\left(\frac{2}{x}\right) = \theta$$

$$2 \tan \theta - \theta + C$$

$$2 \times \frac{\sqrt{x^2-4}}{2} - \arccos\left(\frac{2}{x}\right) + C$$

$$\int \frac{\sqrt{x^2-4}}{x} dx = \sqrt{x^2-4} - \arccos\left(\frac{2}{x}\right) + C$$



After-Exam Survey

Fill in (● not ✓ or ×) one bubble.

This exam was

easy

☐☐☐

reasonable

☒☐

challenging

☐☐

I felt that the WebAssign helped prepare me for this exam

none

☐☐☐

50%

☐☐

well

☒☐

The time I was given to complete this exam was

too little

☐☐☐

just right

☒☐

a lot

☐☐

The exam required

creativity

☐☐☐☐☒

computation

☐☐

